安装□可□

ONTAP Systems Switches

NetApp
November 22, 2021

This PDF was generated from https://docs.netapp.com/zh-cn/ontap-systems-switches/switch-bes-53248/replace-restrictions.html on November 22, 2021. Always check docs.netapp.com for the latest.

目录

安装 BES-53248 集群交换机的许可证 1
限制和限制— BES-53248 交换机 7
编辑参考配置文件（ RCF ） 9

安装 BES-53248 集群交换机的许可证

BES-53248 集群交换机基本型号已获得 16 个 10GbE 或 25GbE 端口和 2 个 100GbE 端口的许可。可以通过购买更多许可证来添加新端口。

以下许可证可用于 BES-53248 集群交换机：

许可证类型	许可证详细信息
支持的固件版本	SW-BES-53248A1-G1-8P-LIC
Broadcom 8P 10-25 ， 2P40-100 许可证密钥， X190005/R	EFOS 3.4.3.3 及更高版本
SW-BES-53248A1-G1-16P-LIC	Broadcom 16P 10-25 ， 4P40-100 许可证密钥， X190005/R
EFOS 3.4.3.3 及更高版本	SW-BES-53248A1-G1-24P 许可证
Broadcom 24P 10-25 ， 6P40-100 许可证密钥， X190005/R	EFOS 3.4.3.3 及更高版本
SW-BES542400-40-100G-LIC	Broadcom 6 端口 40G100G 许可证密钥， X190005/R
EFOS 3.4.4.6 及更高版本	SW-BES53248-8P-10G25G-LIC
Broadcom 8 端口 10G25G 许可证密钥， X190005/R	EFOS 3.4.4.6 及更高版本
SW-BES53248/16P-1025G-LIC	Broadcom 16 端口 10G25G 许可证密钥， X190005/R
EFOS 3.4.4.6 及更高版本	SW-BES532424-24P-1025G-LIC
Broadcom 24 端口 10G25G 许可证密钥， X190005/R	EFOS 3.4.4.6 及更高版本

步骤

1. 将集群交换机连接到管理网络。
2. 使用 `ping` 命令验证与托管 EFOS ，许可证和 RCF 文件的服务器的连接。

   此示例验证交换机是否已连接到 IP 地址为 172.19.2.1 的服务器：

```
(cs2) # **ping 172.19.2.1**
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. 检查交换机 CS2 上的当前许可证使用情况：

   s如何许可

```
(cs2) # **show license**
Reboot needed..................................... No
Number of active licenses......................... 0

License Index  License Type     Status
-------------- ---------------- -----------

No license file found.
```

4. 安装许可证文件。以下示例使用 HTTP 将许可证文件复制到密钥索引 1 。

   重复此步骤可加载更多许可证并使用不同的密钥索引编号。

```
(cs2) # **copy http://172.19.2.1/tmp/efos/license1.dat nvram:license-key
1**

Mode............................................ HTTP
Set Server IP................................... 172.19.2.1
Path............................................ tmp/efos/
Filename........................................ license1.dat
Data Type....................................... license

Management access will be blocked for the duration of the transfer
Are you sure you want to start? (y/n) **y**

File transfer in progress. Management access will be blocked for the
duration of the transfer. Please wait...

License Key transfer operation completed successfully. System reboot is
required.
```

5. 显示所有当前许可证信息，并在重新启动交换机 CS2 之前记下许可证状态：

   s如何许可

```
(cs2) # **show license**

Reboot needed..................................... Yes
Number of active licenses....................... 0


License Index  License Type      Status
-------------- ----------------- --------------------------------
1              Port              License valid but not applied
```

6. 显示所有已许可的端口：

   s如何全部端口 | 排除断开

   只有在重新启动交换机后，才会显示其他许可证文件中的端口。

```
(cs2) # **show port all \| exclude Detach**

                 Admin     Physical   Physical   Link   Link    LACP
Actor
Intf      Type   Mode      Mode       Status     Status Trap    Mode
Timeout
--------- ------ --------- ---------- ---------- ------ ------- ------
--------
0/1              Disable   Auto                  Down   Enable  Enable
long
0/2              Disable   Auto                  Down   Enable  Enable
long
0/3              Disable   Auto                  Down   Enable  Enable
long
0/4              Disable   Auto                  Down   Enable  Enable
long
0/5              Disable   Auto                  Down   Enable  Enable
long
0/6              Disable   Auto                  Down   Enable  Enable
long
0/7              Disable   Auto                  Down   Enable  Enable
long
0/8              Disable   Auto                  Down   Enable  Enable
long
0/9              Disable   Auto                  Down   Enable  Enable
long
0/10             Disable   Auto                  Down   Enable  Enable
long
0/11             Disable   Auto                  Down   Enable  Enable
long
0/12             Disable   Auto                  Down   Enable  Enable
long
0/13             Disable   Auto                  Down   Enable  Enable
long
0/14             Disable   Auto                  Down   Enable  Enable
long
0/15             Disable   Auto                  Down   Enable  Enable
long
0/16             Disable   Auto                  Down   Enable  Enable
long
0/55             Disable   Auto                  Down   Enable  Enable
long
0/56             Disable   Auto                  Down   Enable  Enable
long
```

7. 重新启动交换机：

   re负载

```
(cs2) # **reload**

The system has unsaved changes.
Would you like to save them now? (y/n) **y**

Config file 'startup-config' created successfully .

Configuration Saved!
Are you sure you would like to reset the system? (y/n) **y**
```

8. 检查新许可证是否处于活动状态，并注意此许可证是否已应用：

   s如何许可

```
(cs2) # **show license**

Reboot needed..................................... No
Number of installed licenses.................... 1
Total Downlink Ports enabled.................... 16
Total Uplink Ports enabled...................... 8

License Index  License Type              Status
-------------- -------------------------
------------------------------------
1              Port                      License applied
(cs2) #
```

9. 检查所有新端口是否可用：

   s如何全部端口 | 排除断开

```
(cs2) # **show port all \| exclude Detach**

                  Admin     Physical   Physical   Link   Link    LACP
Actor
Intf      Type   Mode      Mode       Status     Status Trap    Mode
Timeout
--------- ------ --------- ---------- ---------- ------ ------- ------
--------
0/1              Disable    Auto                 Down   Enable  Enable
```

```
long
0/2               Disable    Auto                 Down   Enable  Enable
long
0/3               Disable    Auto                 Down   Enable  Enable
long
0/4               Disable    Auto                 Down   Enable  Enable
long
0/5               Disable    Auto                 Down   Enable  Enable
long
0/6               Disable    Auto                 Down   Enable  Enable
long
0/7               Disable    Auto                 Down   Enable  Enable
long
0/8               Disable    Auto                 Down   Enable  Enable
long
0/9               Disable    Auto                 Down   Enable  Enable
long
0/10              Disable    Auto                 Down   Enable  Enable
long
0/11              Disable    Auto                 Down   Enable  Enable
long
0/12              Disable    Auto                 Down   Enable  Enable
long
0/13              Disable    Auto                 Down   Enable  Enable
long
0/14              Disable    Auto                 Down   Enable  Enable
long
0/15              Disable    Auto                 Down   Enable  Enable
long
0/16              Disable    Auto                 Down   Enable  Enable
long
0/49              Disable   100G Full             Down   Enable  Enable
long
0/50              Disable   100G Full             Down   Enable  Enable
long
0/51              Disable   100G Full             Down   Enable  Enable
long
0/52              Disable   100G Full             Down   Enable  Enable
long
0/53              Disable   100G Full             Down   Enable  Enable
long
0/54              Disable   100G Full             Down   Enable  Enable
long
0/55              Disable   100G Full             Down   Enable  Enable
long
0/56              Disable   100G Full             Down   Enable  Enable
```

```
long

(cs2) #
```

限制和限制— BES-53248 交换机

如果在安装许可证时出现问题，应先运行以下调试命令，然后再运行 `copy` 命令来安装许可证。

要使用的调试命令：

`debug transferdebug license`

```
(cs2) # debug transfer
Debug transfer output is enabled.
(cs2) # debug license
Enabled capability licensing debugging.
```

在启用 `debug transfer` 和 `debug license` 选项的情况下运行 `copy` 命令时，将返回以下日志输出：

```
transfer.c(3083):Transfer process  key or certificate file type = 43
transfer.c(3229):Transfer process  key/certificate cmd = cp
/mnt/download//license.dat.1 /mnt/fastpath/ >/dev/null 2>&1CAPABILITY
LICENSING :
Fri Sep 11 13:41:32 2020: License file with index 1 added.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Validating hash value
29de5e9a8af3e510f1f16764a13e8273922d3537d3f13c9c3d445c72a180a2e6.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Parsing JSON buffer {
  "license": {
    "header": {
      "version": "1.0",
      "license-key": "964B-2D37-4E52-BA14",
      "serial-number": "QTFCU38290012",
      "model": "BES-53248"
  },
  "description": "",
  "ports": "0+6"
  }
}.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: License data does not
contain 'features' field.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Serial number
QTFCU38290012 matched.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Model BES-53248 matched.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Feature not found in
license file with index = 1.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Applying license file 1.
```

在调试输出中检查以下内容：

- 检查序列号是否匹配： `s`序列号 `QTCU38290012` 匹配。
- 检查交换机型号是否匹配： `model BES-53248 matched` 。
- 检查指定的许可证索引是否以前未使用过。如果已使用许可证索引，则返回以下错误： `License file /mnt/download//license.date.1 already exists.`
- 端口许可证不是功能许可证。因此，应遵循以下语句：在索引为 1.` 的许可证文件中未找到 `功能

使用 `copy` 命令将端口许可证备份到服务器：

```
(cs2) # copy nvram:license-key 1
scp://<UserName>@<IP_address>/saved_license_1.dat
```

请参见 "为 BES-53248 集群交换机安装许可证" 有关可用许可证支持的固件版本的详细信息。

如果需要将交换机软件从 3.4.4.6 版降级，则许可证将被删除。这是预期行为。

在还原到软件的旧版本之前，您必须安装相应的旧版许可证。

编辑参考配置文件（RCF）

要激活新许可的端口，您需要编辑最新版本的 RCF 并取消注释适用的端口详细信息。默认许可证会激活端口 0/1 到 0/16 以及从 0/55 到 0/56 ，而新许可的端口将位于端口 0/17 到 0/54 之间，具体取决于可用许可证的类型和数量。

如果您尝试编辑先前安装的 RCF ，则此过程可能会失败，因为 RCF 中的其他区域已有配置，请参见 [Edit a previously installed RCF file]。

有关可在 BES-53248 集群交换机上使用的许可证类型的详细信息，请参见 "为 BES-53248 集群交换机安装许可证"。

例如，要激活 SW-BES54241-40-100G-LIC 许可证，必须取消注释 RCF 中的以下部分：

```
.
.
!
! 2-port or 6-port 40/100GbE node port license block
!
interface 0/49
no shutdown
description "40/100GbE Node Port"
!speed 100G full-duplex
speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/50
no shutdown
description "40/100GbE Node Port"
!speed 100G full-duplex
speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
```

```
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/51
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/52
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/53
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
```

```
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/54
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
.
.
```

> 对于介于 0/49 到 0/54 之间的高速端口，取消对每个端口的注释，但在 RCF 中仅对其中每个端口的一条 * 速度 * 线取消注释，可以：

- 速度 100G 全双工
- 速度 40G 全双工

如示例所示。

对于介于 0/17 到 0/48 之间的低速度端口，激活相应的许可证后，取消对整个 8 端口部分的注释。

编辑先前安装的 RCF 文件

编辑先前安装的 RCF 文件并运行 `sscript apply` 命令后，可能会收到以下错误消息：

```
(CS1) #script apply BES-53248_RCF_v1.6-Cluster-HA.scr
 Are you sure you want to apply the configuration script? (y/n)
```

选择 `y` 后，您将收到以下错误消息：

```
config
 ...
 match cos 5
 Unrecognized command : match cos 5
 Error! in configuration script file at line number 40.
 CLI Command :: match cos 5.
 Aborting script.
```

要避免或解决此问题描述，您可以选择以下选项之一：

- 要避免此错误，您可以使用以下操作步骤：
  a. 再创建一个仅包含新端口配置的 RCF 文件。
  b. 将第二个 RCF 文件复制到交换机。
  c. 将此脚本应用于交换机。

     s记录应用

- 要解决此错误，请参见知识库文章： "error在应用新 RCF 时，配置脚本文件的行号为 XX"

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.